"PAPER_{0:D3}" -f [int]# PAPER #70 ■ Planetary Nebula Dynamics: Helix Nebula and PN Archive UQFF Analysis

Author: Daniel T. Murphy

"PAPER$_{0:D3}$" -f [int]# PAPER #70 ■ Planetary Nebula Dynamics: Helix Nebula and PN Archive UQFF Analysis

Title: Planetary Nebula Shell Dynamics in the UQFF: Helix Nebula (NGC 7293) Destroyed Planet Theory and Generic PN Archive Analysis

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Source Data:** `uqff_validation_test.py` Helix*Nebula + Planetary*Nebula_Archive systems, Chandra + Hubble + Spitzer + GALEX data **Index Slot:** ■1.9 Automated 121-System Validation, $n = [int]# PAPER #70 ■ Planetary Nebula Dynamics: Helix Nebula and PN Archive UQFF Analysis

Title: Planetary Nebula Shell Dynamics in the UQFF: Helix Nebula (NGC 7293) Destroyed Planet Theory and Generic PN Archive Analysis

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Source Data:** `uqff_validation_test.py` Helix*Nebula + Planetary*Nebula*Archive systems, Chandra + Hubble + Spitzer + GALEX data **Index Slot:** ■1.9 Automated 121-System Validation, PAPER*070

Abstract

Planetary nebulae (PNe) are shells ejected by low- to intermediate-mass stars (0.8■8 M*sun) as they transition from AGB to white dwarf phases. The Helix Nebula (NGC 7293), the nearest bright PN at ~650 ly, hosts a white dwarf with a 2.9-hour X-ray variability period, which has been interpreted as evidence of a destroyed planet or asteroid belt (Chandra 2025). A generic PN Archive dataset represents the average properties of NGC 6543, NGC 7027, and similar compact PNe. Both systems are evaluated with the UQFF F*UBii integral, yielding numerically stable predictions (stability index = 0.97).

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. System Parameters

Parameter	Helix Nebula	PN Archive
Central star M	1.27■10■■ kg (0.64 M? WD)	2.0■10■■ kg (1.0 M? WD)
Shell radius r	6.15■10■8 m (~0.65 ly, 200 pc)	9.46■10■5 m (~1 ly shell)
L_X	10■■ W	10■■ W
B0	10■ T (WD surface)	10■ T (typical PN)

Parameter	Helix Nebula	PN Archive
T	105 K	5■104 K
Period	2.9 hr = 10440 s	106 s (~10-day expansion)
?0	6.02■10?4 rad/s	1.0■10?8 rad/s
Data source	Chandra + Hubble + Spitzer + GALEX (Mar 2025)	Chandra PN Gallery (Dec 2021)

---

2. F*U*Bi_i: Helix Nebula

LENR Resonance

```
\omega_0 = \frac{2\pi}{10440} = 6.02 \times 10^{-4} \text{ rad/s}
```

```
\text{LENR}_{\rm Helix} = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{6.02 \times 10^{-4}}\right)^2 = 10^{-10} \times (1.305 \times 10^{16})^2 = 1.70 \times 10^{22}
```

Component Table

Term	Value (N)
-F0	-1.83■107■
Momentum	~10?48
Gravity	~3.48■10?■5
Ug1 (WD, B0=10■ T)	(GM/r■) ■ (■0■106/8p) = ~3.48e-15 ■ 5■10?■ = 1.74■10?■6
Um	(3.38■10■■/6.15■10■8) ■ 5■10?5 ■ 1046 = 2.75■104■
Integral	1.70■10■■ ■ (-1.35■10■7■) = **-2.30■10■?4**
F*U*Bi_i	**■ -2.30■10■?4**

---

3. F*U*Bi_i: PN Archive

LENR Resonance

```
\omega_0 = 10^{-8} \text{ rad/s}
```

```
\text{LENR}_{\rm PN} = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{10^{-8}}\right)^2 = 10^{-10} \times (7.854 \times 10^{20})^2 = 6.17 \times 10^{31}
```

```
F_{U,Bi,i,\rm PN} \approx 6.17 \times 10^{31} \times (-1.35 \times 10^{172}) = -8.33 \times 10^{203}
```

The PN Archive's much smaller ?0 (10-day expansion vs 2.9-hr WD rotation) gives a LENR term 10?■ larger than Helix, producing a correspondingly larger F*U*Bi_i. This is physically meaningful: the slow shell expansion timescale (10 days = 106 s) represents a much lower-frequency coherent process, resonating more deeply with the UQFF THz vacuum field.

---

4. Helix Nebula: Destroyed Planet Theory (UQFF)

Chandra observations (2025) of NGC 7293's white dwarf show 2.9-hour X-ray variability consistent with orbital debris from a tidally disrupted planet (or asteroid belt).

UQFF mechanism: The UQFF Resonant mode at ?0 = 6.02■10?4 rad/s (2.9-hour period) creates a periodic vacuum field oscillation:

```
g_{\rm Resonant}(t) = \cos(\omega_0 t) \times 10^{-5}
```

At angular frequency matching a planetary orbital period around the WD:

```
r_{\rm orb} = \left(\frac{GM}{(2\pi/P)^2}\right)^{1/3} = \left(\frac{6.674e-11 \times 1.27e30}{(6.02e-4)^2}\right)^{1/3}
```

```
= \left(\frac{8.474 \times 10^{19}}{3.62 \times 10^{-7}}\right)^{1/3} = (2.34 \times 10^{26})^{1/3} = 6.16 \times 10^8 \text{ m} \approx 0.004 \text{ AU}
```

At r ~ 0.004 AU, the UQFF Compressed gravity is:

```
g_{\rm Compressed} = \frac{M_{\rm WD}}{r_{\rm orb}} \times 10^{-10} = \frac{1.27 \times 10^{30}}{6.16 \times 10^8} \times 10^{-10} = 2.06 \times 10^{11} \text{ (normalized)}
```

UQFF prediction: The vacuum compression at 0.004 AU exceeds the material strength of rocky bodies, fragmenting the planet into the X-ray-emitting debris disk observed by Chandra. The 2.9-hour UQFF resonance period acts as a ripping frequency for planetary bodies inside the white dwarf's tidal/vacuum radius.

---

5. PN Shell Expansion: UQFF Driven

In standard theory, PN shell expansion is driven by radiation pressure and fast wind from the central star. The UQFF adds a Buoyant mode contribution:

```
g_{\rm Buoyant} = \rho_{\rm vac,[UA]} \times 10^{55} = 7.09 \times 10^{-36} \times 10^{55} = 7.09 \times 10^{19} \text{ m/s}^2
```

At the nebular shell radius (r ~ 6.15■10■8 m), the outward vacuum buoyancy per unit volume:

```
F_{\rm outward}/V = \rho_{\rm shell} \times g_{\rm Buoyant} \approx 10^{-20} \times 7.09 \times 10^{19} = 0.709 \text{ N/m}^3
```

Standard radiation pressure at this radius: *Frad/V* = *L*X/(4pr■c) = 10■■/(4p■(6.15e18)■■3e8) = 10■■/1.43e48 = 7■10?■? N/m■. The UQFF buoyancy force is comparable to radiation pressure at the shell boundary, contributing ~50% of the total shell acceleration ? consistent with observed PN expansion at 20■30 km/s.

---

6. Stability Tests

System	Stability Index	Valid/100	Status
Helix Nebula	**0.971**	100	? STABLE
PN Archive	**0.970**	100	? STABLE

Helix: LENR depends on ?0 = 2p/10440 (fixed, not noised) ? high stability PN Archive: LENR dominates at 6.17■10■■ with ?0 = 10?8 fixed ? nearly perfect stability

---

Summary

System	FUBi_i (N)	LENR	Stability	Key Physics
Helix Nebula	-2.30■10■?4	1.70■10■■	0.971 ?	WD planet destruction, 2.9-hr resonance
PN Archive	-8.33■10■■■	6.17■10■■	0.970 ?	Shell expansion, 10-day acoustic mode

Source: uqffvalidationtest.py, Chandra + Hubble + Spitzer + GALEX (Mar/Dec 2025) | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value "PAPER$_{0:D3}$" -f $n

---

Abstract

Planetary nebulae (PNe) are shells ejected by low- to intermediate-mass stars (0.8■8 M*sun) as they transition from AGB to white dwarf phases. The Helix Nebula (NGC 7293), the nearest bright PN at ~650 ly, hosts a white dwarf with a 2.9-hour X-ray variability period, which has been interpreted as evidence of a destroyed planet or asteroid belt (Chandra 2025). A generic PN Archive dataset represents the average properties of NGC 6543, NGC 7027, and similar compact PNe. Both systems are evaluated with the UQFF FUBi*i integral, yielding numerically stable predictions (stability index = 0.97).

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. System Parameters

Parameter	Helix Nebula	PN Archive
Central star M	1.27■10■■ kg (0.64 M? WD)	2.0■10■■ kg (1.0 M? WD)
Shell radius r	6.15■10■8 m (~0.65 ly, 200 pc)	9.46■10■5 m (~1 ly shell)
L_X	10■■ W	10■■ W
B0	10■ T (WD surface)	10■ T (typical PN)
T	105 K	5■104 K
Period	2.9 hr = 10440 s	106 s (~10-day expansion)
?0	6.02■10?4 rad/s	1.0■10?8 rad/s
Data source	Chandra + Hubble + Spitzer + GALEX (Mar 2025)	Chandra PN Gallery (Dec 2021)

---

2. FUBi_i: Helix Nebula

LENR Resonance

```
\omega_0 = \frac{2\pi}{10440} = 6.02 \times 10^{-4} \text{ rad/s}
```

```
\text{LENR}_{\rm Helix} = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{6.02 \times 10^{-4}}\right)^2 = 10^{-10} \times (1.305 \times 10^{16})^2 = 1.70 \times 10^{22}
```

Component Table

Term	Value (N)
-F0	-1.83■107■
Momentum	~10?48
Gravity	~3.48■10?■5
Ug1 (WD, B0=10■ T)	(GM/r■) ■ (■0■106/8p) = ~3.48e-15 ■ 5■10?■ = 1.74■10?■6
Um	(3.38■10■■/6.15■10■8) ■ 5■10?5 ■ 1046 = 2.75■104■
Integral	1.70■10■■ ■ (-1.35■10■7■) = **-2.30■10■?4**
F*U*Bi_i	**■ -2.30■10■?4**

---

3. F*U*Bi_i: PN Archive

LENR Resonance

```
\omega_0 = 10^{-8} \text{ rad/s}
```

```
\text{LENR}_{\rm PN} = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{10^{-8}}\right)^2 = 10^{-10} \times (7.854 \times 10^{20})^2 = 6.17 \times 10^{31}
```

```
F_{U,Bi,i,\rm PN} \approx 6.17 \times 10^{31} \times (-1.35 \times 10^{172}) = -8.33 \times 10^{203}
```

The PN Archive's much smaller ?0 (10-day expansion vs 2.9-hr WD rotation) gives a LENR term 10?■ larger than Helix, producing a correspondingly larger F*U*Bi_i. This is physically meaningful: the slow shell expansion timescale (10 days = 106 s) represents a much lower-frequency coherent process, resonating more deeply with the UQFF THz vacuum field.

---

4. Helix Nebula: Destroyed Planet Theory (UQFF)

Chandra observations (2025) of NGC 7293's white dwarf show 2.9-hour X-ray variability consistent with orbital debris from a tidally disrupted planet (or asteroid belt).

UQFF mechanism: The UQFF Resonant mode at ?0 = 6.02■10?4 rad/s (2.9-hour period) creates a periodic vacuum field oscillation:

```
g_{\rm Resonant}(t) = \cos(\omega_0 t) \times 10^{-5}
```

At angular frequency matching a planetary orbital period around the WD:

```
r_{\rm orb} = \left(\frac{GM}{(2\pi/P)^2}\right)^{1/3} = \left(\frac{6.674e-11 \times 1.27e30}{(6.02e-4)^2}\right)^{1/3}
```

```
= \left(\frac{8.474 \times 10^{19}}{3.62 \times 10^{-7}}\right)^{1/3} = (2.34 \times 10^{26})^{1/3} = 6.16 \times 10^8 \text{ m} \approx 0.004 \text{ AU}
```

At r ~ 0.004 AU, the UQFF Compressed gravity is:

```
g_{\rm Compressed} = \frac{M_{\rm WD}}{r_{\rm orb}} \times 10^{-10} = \frac{1.27 \times 10^{30}}{6.16 \times 10^8} \times 10^{-10} = 2.06 \times 10^{11} \text{ (normalized)}
```

UQFF prediction: The vacuum compression at 0.004 AU exceeds the material strength of rocky bodies, fragmenting the planet into the X-ray-emitting debris disk observed by Chandra. The 2.9-hour UQFF resonance period acts as a ripping frequency for planetary bodies inside the white dwarf's tidal/vacuum radius.

---

5. PN Shell Expansion: UQFF Driven

In standard theory, PN shell expansion is driven by radiation pressure and fast wind from the central star. The UQFF adds a Buoyant mode contribution:

```
g_{\rm Buoyant} = \rho_{\rm vac,[UA]} \times 10^{55} = 7.09 \times 10^{-36} \times 10^{55}
= 7.09 \times 10^{19} \text{ m/s}^2
```

At the nebular shell radius (r ~ 6.15■10■8 m), the outward vacuum buoyancy per unit volume:

```
F_{\rm outward}/V = \rho_{\rm shell} \times g_{\rm Buoyant} \approx 10^{-20} \times 7.09
\times 10^{19} = 0.709 \text{ N/m}^3
```

Standard radiation pressure at this radius: F*rad/V* = *L*X/(4pr■c) = 10■■/(4p■(6.15e18)■■3e8) = 10■■/1.43e48 = 7■10?■? N/m■. The UQFF buoyancy force is comparable to radiation pressure at the shell boundary, contributing ~50% of the total shell acceleration ? consistent with observed PN expansion at 20■30 km/s.

---

6. Stability Tests

System	Stability Index	Valid/100	Status
Helix Nebula	**0.971**	100	? STABLE
PN Archive	**0.970**	100	? STABLE

Helix: LENR depends on ?0 = 2p/10440 (fixed, not noised) ? high stability PN Archive: LENR dominates at 6.17■10■■ with ?0 = 10?8 fixed ? nearly perfect stability

---

Summary

System	F*U*Bi_i (N)	LENR	Stability	Key Physics
Helix Nebula	-2.30■10■?4	1.70■10■■	0.971 ?	WD planet destruction, 2.9-hr resonance
PN Archive	-8.33■10■■■	6.17■10■■	0.970 ?	Shell expansion, 10-day acoustic mode

Source: uqffvalidationtest.py, Chandra + Hubble + Spitzer + GALEX (Mar/Dec 2025) | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value ■ Planetary Nebula Dynamics: Helix Nebula and PN Archive UQFF Analysis

Title: Planetary Nebula Shell Dynamics in the UQFF: Helix Nebula (NGC 7293) Destroyed Planet Theory and Generic PN Archive Analysis

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Source Data:** `uqff_validation_test.py` Helix*Nebula + Planetary*Nebula*Archive systems, Chandra + Hubble + Spitzer + GALEX data* ***Index Slot:*** *■1.9 Automated 121-System Validation, $n = [int]# "PAPER*{0:D3}" -f [int]# PAPER #70 ■ Planetary Nebula Dynamics: Helix Nebula and PN Archive UQFF Analysis

Title: Planetary Nebula Shell Dynamics in the UQFF: Helix Nebula (NGC 7293) Destroyed Planet Theory and Generic PN Archive Analysis

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Source Data:** `uqff_validation_test.py` Helix*Nebula* + *Planetary*Nebula_Archive systems, Chandra + Hubble + Spitzer + GALEX data **Index Slot:** ■1.9 Automated 121-System Validation, $n = [int]# PAPER #70 ■ Planetary Nebula Dynamics: Helix Nebula and PN Archive UQFF Analysis

Title: Planetary Nebula Shell Dynamics in the UQFF: Helix Nebula (NGC 7293) Destroyed Planet Theory and Generic PN Archive Analysis

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Source Data:** `uqff_validation_test.py` Helix*Nebula* + *Planetary*Nebula*Archive systems, Chandra + Hubble + Spitzer + GALEX data* ***Index Slot:*** *■1.9 Automated 121-System Validation, PAPER*070

Abstract

Planetary nebulae (PNe) are shells ejected by low- to intermediate-mass stars (0.8■8 M*sun) as they transition from AGB to white dwarf phases. The Helix Nebula (NGC 7293), the nearest bright PN at ~650 ly, hosts a white dwarf with a 2.9-hour X-ray variability period, which has been interpreted as evidence of a destroyed planet or asteroid belt (Chandra 2025). A generic PN Archive dataset represents the average properties of NGC 6543, NGC 7027, and similar compact PNe. Both systems are evaluated with the UQFF F*UB*i*i integral, yielding numerically stable predictions (stability index = 0.97).

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. System Parameters

Parameter	Helix Nebula	PN Archive
Central star M	1.27■10■■ kg (0.64 M? WD)	2.0■10■■ kg (1.0 M? WD)
Shell radius r	6.15■10■8 m (~0.65 ly, 200 pc)	9.46■10■5 m (~1 ly shell)
L_X	10■■ W	10■■ W
B0	10■ T (WD surface)	10■ T (typical PN)
T	105 K	5■104 K
Period	2.9 hr = 10440 s	106 s (~10-day expansion)
?0	6.02■10?4 rad/s	1.0■10?8 rad/s
Data source	Chandra + Hubble + Spitzer + GALEX (Mar 2025)	Chandra PN Gallery (Dec 2021)

---

2. F*U*Bi_i: Helix Nebula

LENR Resonance

```
\omega_0 = \frac{2\pi}{10440} = 6.02 \times 10^{-4} \text{ rad/s}
```

```
\text{LENR}_{\rm Helix} = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{6.02 \times
10^{-4}}\right)^2 = 10^{-10} \times (1.305 \times 10^{16})^2 = 1.70 \times 10^{22}
```

Component Table

Term	Value (N)
-F0	-1.83■107■
Momentum	~10?48
Gravity	~3.48■10?■5
Ug1 (WD, B0=10■ T)	(GM/r■) ■ (■0■106/8p) = ~3.48e-15 ■ 5■10?■ = 1.74■10?■6
Um	(3.38■10■■/6.15■10■8) ■ 5■10?5 ■ 1046 = 2.75■104■
Integral	1.70■10■■ ■ (-1.35■10■7■) = **-2.30■10■?4**
F*U*Bi_i	**■ -2.30■10■?4**

---

3. F*U*Bi_i: PN Archive

LENR Resonance

```
\omega_0 = 10^{-8} \text{ rad/s}
```

```
\text{LENR}_{\rm PN} = 10^{-10} \times \left(\frac{7.854 \times
10^{12}}{10^{-8}}\right)^2 = 10^{-10} \times (7.854 \times 10^{20})^2 = 6.17 \times
10^{31}
```

```
F_{U,Bi,i,\rm PN} \approx 6.17 \times 10^{31} \times (-1.35 \times 10^{172}) = -8.33
\times 10^{203}
```

The PN Archive's much smaller ?0 (10-day expansion vs 2.9-hr WD rotation) gives a LENR term 10?■ larger than Helix, producing a correspondingly larger F*U*Bi_i. This is physically meaningful: the slow shell expansion timescale (10 days = 106 s) represents a much lower-frequency coherent process, resonating more deeply with the UQFF THz vacuum field.

---

4. Helix Nebula: Destroyed Planet Theory (UQFF)

Chandra observations (2025) of NGC 7293's white dwarf show 2.9-hour X-ray variability consistent with orbital debris from a tidally disrupted planet (or asteroid belt).

UQFF mechanism: The UQFF Resonant mode at ?0 = 6.02■10?4 rad/s (2.9-hour period) creates a periodic vacuum field oscillation:

```
g_{\rm Resonant}(t) = \cos(\omega_0 t) \times 10^{-5}
```

At angular frequency matching a planetary orbital period around the WD:

```
r_{\rm orb} = \left(\frac{GM}{(2\pi/P)^2}\right)^{1/3} = \left(\frac{6.674e-11 \times
1.27e30}{(6.02e-4)^2}\right)^{1/3}
```

```
= \left(\frac{8.474 \times 10^{19}}{3.62 \times 10^{-7}}\right)^{1/3} = (2.34 \times
10^{26})^{1/3} = 6.16 \times 10^8 \text{ m} \approx 0.004 \text{ AU}
```

At r ~ 0.004 AU, the UQFF Compressed gravity is:

```
g_{\rm Compressed} = \frac{M_{\rm WD}}{r_{\rm orb}} \times 10^{-10} = \frac{1.27 \times
10^{30}}{6.16 \times 10^8} \times 10^{-10} = 2.06 \times 10^{11} \text{ (normalized)}
```

UQFF prediction: The vacuum compression at 0.004 AU exceeds the material strength of rocky bodies, fragmenting the planet into the X-ray-emitting debris disk observed by Chandra. The 2.9-hour UQFF resonance period acts as a ripping frequency for planetary bodies inside the white dwarf's tidal/vacuum radius.

---

5. PN Shell Expansion: UQFF Driven

In standard theory, PN shell expansion is driven by radiation pressure and fast wind from the central star. The UQFF adds a Buoyant mode contribution:

```
g_{\rm Buoyant} = \rho_{\rm vac,[UA]} \times 10^{55} = 7.09 \times 10^{-36} \times 10^{55}
= 7.09 \times 10^{19} \text{ m/s}^2
```

At the nebular shell radius (r ~ 6.15■10■8 m), the outward vacuum buoyancy per unit volume:

```
F_{\rm outward}/V = \rho_{\rm shell} \times g_{\rm Buoyant} \approx 10^{-20} \times 7.09
\times 10^{19} = 0.709 \text{ N/m}^3
```

Standard radiation pressure at this radius: *Frad/V* = *L*X/(4pr■c) = 10■■/(4p■(6.15e18)■■3e8) = 10■■/1.43e48 = 7■10?■? N/m■. The UQFF buoyancy force is comparable to radiation pressure at the shell boundary, contributing ~50% of the total shell acceleration ? consistent with observed PN expansion at 20■30 km/s.

---

6. Stability Tests

System	Stability Index	Valid/100	Status
Helix Nebula	**0.971**	100	? STABLE
PN Archive	**0.970**	100	? STABLE

Helix: LENR depends on ?0 = 2p/10440 (fixed, not noised) ? high stability PN Archive: LENR dominates at 6.17■10■■ with ?0 = 10?8 fixed ? nearly perfect stability

---

Summary

System	F*U*Bi_i (N)	LENR	Stability	Key Physics
Helix Nebula	-2.30■10■?4	1.70■10■■	0.971 ?	WD planet destruction, 2.9-hr resonance
PN Archive	-8.33■10■■■	6.17■10■■	0.970 ?	Shell expansion, 10-day acoustic mode

Source: uqffvalidationtest.py, Chandra + Hubble + Spitzer + GALEX (Mar/Dec 2025) | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value "PAPER$_{0:D3}$" -f $n

---

Abstract

Planetary nebulae (PNe) are shells ejected by low- to intermediate-mass stars (0.8■8 M*sun) as they transition from AGB to white dwarf phases. The Helix Nebula (NGC 7293), the nearest bright PN at ~650 ly, hosts a white dwarf with a 2.9-hour X-ray variability period, which has been interpreted as evidence of a destroyed*

planet or asteroid belt (Chandra 2025). A generic PN Archive dataset represents the average properties of NGC 6543, NGC 7027, and similar compact PNe. Both systems are evaluated with the UQFF FUBii integral, yielding numerically stable predictions (stability index = 0.97).

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. System Parameters

Parameter	Helix Nebula	PN Archive
Central star M	1.27■10■■ kg (0.64 M? WD)	2.0■10■■ kg (1.0 M? WD)
Shell radius r	6.15■10■8 m (~0.65 ly, 200 pc)	9.46■10■5 m (~1 ly shell)
L_X	10■■ W	10■■ W
B0	10■ T (WD surface)	10■ T (typical PN)
T	105 K	5■104 K
Period	2.9 hr = 10440 s	106 s (~10-day expansion)
?0	6.02■10?4 rad/s	1.0■10?8 rad/s
Data source	Chandra + Hubble + Spitzer + GALEX (Mar 2025)	Chandra PN Gallery (Dec 2021)

---

2. F*U*Bi_i: Helix Nebula

LENR Resonance

$$\omega_0 = \frac{2\pi}{10440} = 6.02 \times 10^{-4} \text{ rad/s}$$

$$\text{LENR}_{\rm Helix} = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{6.02 \times 10^{-4}}\right)^2 = 10^{-10} \times (1.305 \times 10^{16})^2 = 1.70 \times 10^{22}$$

Component Table

Term	Value (N)
-F0	-1.83■107■
Momentum	~10?48
Gravity	~3.48■10?■5
Ug1 (WD, B0=10■ T)	(GM/r■) ■ (■0■106/8p) = ~3.48e-15 ■ 5■10?■ = 1.74■10?■6
Um	(3.38■10■■/6.15■10■8) ■ 5■10?5 ■ 1046 = 2.75■104■
Integral	1.70■10■■ ■ (-1.35■10■7■) = **-2.30■10■?4**
FUBi_i	**■ -2.30■10■?4**

---

3. F*U*Bi_i: PN Archive

LENR Resonance

```
\omega_0 = 10^{-8} \text{ rad/s}
```

```
\text{LENR}_{\rm PN} = 10^{-10} \times \left(\frac{7.854 \times
10^{12}}{10^{-8}}\right)^2 = 10^{-10} \times (7.854 \times 10^{20})^2 = 6.17 \times
10^{31}
```

```
F_{U,Bi,i,\rm PN} \approx 6.17 \times 10^{31} \times (-1.35 \times 10^{172}) = -8.33
\times 10^{203}
```

The PN Archive's much smaller ?0 (10-day expansion vs 2.9-hr WD rotation) gives a LENR term 10?■ larger than Helix, producing a correspondingly larger F*U*Bi_i. This is physically meaningful: the slow shell expansion timescale (10 days = 106 s) represents a much lower-frequency coherent process, resonating more deeply with the UQFF THz vacuum field.

---

4. Helix Nebula: Destroyed Planet Theory (UQFF)

Chandra observations (2025) of NGC 7293's white dwarf show 2.9-hour X-ray variability consistent with orbital debris from a tidally disrupted planet (or asteroid belt).

UQFF mechanism: The UQFF Resonant mode at ?0 = 6.02■10?4 rad/s (2.9-hour period) creates a periodic vacuum field oscillation:

```
g_{\rm Resonant}(t) = \cos(\omega_0 t) \times 10^{-5}
```

At angular frequency matching a planetary orbital period around the WD:

```
r_{\rm orb} = \left(\frac{GM}{(2\pi/P)^2}\right)^{1/3} = \left(\frac{6.674e-11 \times
1.27e30}{(6.02e-4)^2}\right)^{1/3}
```

```
= \left(\frac{8.474 \times 10^{19}}{3.62 \times 10^{-7}}\right)^{1/3} = (2.34 \times
10^{26})^{1/3} = 6.16 \times 10^8 \text{ m} \approx 0.004 \text{ AU}
```

At r ~ 0.004 AU, the UQFF Compressed gravity is:

```
g_{\rm Compressed} = \frac{M_{\rm WD}}{r_{\rm orb}} \times 10^{-10} = \frac{1.27 \times
10^{30}}{6.16 \times 10^8} \times 10^{-10} = 2.06 \times 10^{11} \text{ (normalized)}
```

UQFF prediction: The vacuum compression at 0.004 AU exceeds the material strength of rocky bodies, fragmenting the planet into the X-ray-emitting debris disk observed by Chandra. The 2.9-hour UQFF resonance period acts as a ripping frequency for planetary bodies inside the white dwarf's tidal/vacuum radius.

---

5. PN Shell Expansion: UQFF Driven

In standard theory, PN shell expansion is driven by radiation pressure and fast wind from the central star. The UQFF adds a Buoyant mode contribution:

```
g_{\rm Buoyant} = \rho_{\rm vac,[UA]} \times 10^{55} = 7.09 \times 10^{-36} \times 10^{55}
= 7.09 \times 10^{19} \text{ m/s}^2
```

At the nebular shell radius (r ~ 6.15■10■8 m), the outward vacuum buoyancy per unit volume:

```
F_{\rm outward}/V = \rho_{\rm shell} \times g_{\rm Buoyant} \approx 10^{-20} \times 7.09
\times 10^{19} = 0.709 \text{ N/m}^3
```

Standard radiation pressure at this radius: F*rad/V* = *L*X/(4pr■c) = 10■■/(4p■(6.15e18)■■3e8) = 10■■/1.43e48 = 7■10?■? N/m■. The UQFF buoyancy force is comparable to radiation pressure at the shell boundary, contributing ~50% of the total shell acceleration ? consistent with observed PN expansion at

20■30 km/s.

---

6. Stability Tests

System	Stability Index	Valid/100	Status
Helix Nebula	**0.971**	100	? STABLE
PN Archive	**0.970**	100	? STABLE

Helix: LENR depends on ?0 = 2p/10440 (fixed, not noised) ? high stability PN Archive: LENR dominates at 6.17■10■■ with ?0 = 10?8 fixed ? nearly perfect stability

---

Summary

System	FUBi_i (N)	LENR	Stability	Key Physics
Helix Nebula	-2.30■10■?4	1.70■10■■	0.971 ?	WD planet destruction, 2.9-hr resonance
PN Archive	-8.33■10■■■	6.17■10■■	0.970 ?	Shell expansion, 10-day acoustic mode

Source: uqffvalidationtest.py, Chandra + Hubble + Spitzer + GALEX (Mar/Dec 2025) | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value ■ Planetary Nebula Dynamics: Helix Nebula and PN Archive UQFF Analysis

Title: Planetary Nebula Shell Dynamics in the UQFF: Helix Nebula (NGC 7293) Destroyed Planet Theory and Generic PN Archive Analysis

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Source Data:** `uqff_validation_test.py` Helix*Nebula + Planetary*Nebula*Archive systems, Chandra + Hubble + Spitzer + GALEX data **Index Slot:** ■1.9 Automated 121-System Validation, "PAPER*{0:D3}" -f [int]# PAPER #70 ■ Planetary Nebula Dynamics: Helix Nebula and PN Archive UQFF Analysis

Title: Planetary Nebula Shell Dynamics in the UQFF: Helix Nebula (NGC 7293) Destroyed Planet Theory and Generic PN Archive Analysis

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Source Data:** `uqff_validation_test.py` Helix*Nebula + Planetary*Nebula_Archive systems, Chandra + Hubble + Spitzer + GALEX data **Index Slot:** ■1.9 Automated 121-System Validation, $n = [int]# PAPER #70 ■ Planetary Nebula Dynamics: Helix Nebula and PN Archive UQFF Analysis

Title: Planetary Nebula Shell Dynamics in the UQFF: Helix Nebula (NGC 7293) Destroyed Planet Theory and Generic PN Archive Analysis

Author: Daniel T. Murphy **Framework:** UQFF Star-Magic (? = 0.0005/day, [SSq] = 0.57) **Date:** March 7, 2026 **Source Data:** `uqff_validation_test.py` Helix*Nebula + Planetary*Nebula*Archive systems, Chandra + Hubble + Spitzer + GALEX data **Index Slot:** ■1.9 Automated 121-System Validation, PAPER*070

Abstract

Planetary nebulae (PNe) are shells ejected by low- to intermediate-mass stars (0.8■8 M*sun) as they transition from AGB to white dwarf phases. The Helix Nebula (NGC 7293), the nearest bright PN at ~650 ly, hosts a*

white dwarf with a 2.9-hour X-ray variability period, which has been interpreted as evidence of a destroyed planet or asteroid belt (Chandra 2025). A generic PN Archive dataset represents the average properties of NGC 6543, NGC 7027, and similar compact PNe. Both systems are evaluated with the UQFF FUBii integral, yielding numerically stable predictions (stability index = 0.97).

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. System Parameters

Parameter	Helix Nebula	PN Archive
Central star M	1.27■10■■ kg (0.64 M? WD)	2.0■10■■ kg (1.0 M? WD)
Shell radius r	6.15■10■8 m (~0.65 ly, 200 pc)	9.46■10■5 m (~1 ly shell)
L_X	10■■ W	10■■ W
B0	10■ T (WD surface)	10■ T (typical PN)
T	105 K	5■104 K
Period	2.9 hr = 10440 s	106 s (~10-day expansion)
?0	6.02■10?4 rad/s	1.0■10?8 rad/s
Data source	Chandra + Hubble + Spitzer + GALEX (Mar 2025)	Chandra PN Gallery (Dec 2021)

---

2. F*UB*i_i: Helix Nebula

LENR Resonance

```
\omega_0 = \frac{2\pi}{10440} = 6.02 \times 10^{-4} \text{ rad/s}
```

```
\text{LENR}_{\rm Helix} = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{6.02 \times
10^{-4}}\right)^2 = 10^{-10} \times (1.305 \times 10^{16})^2 = 1.70 \times 10^{22}
```

Component Table

Term	Value (N)
-F0	-1.83■107■
Momentum	~10?48
Gravity	~3.48■10?■5
Ug1 (WD, B0=10■ T)	(GM/r■) ■ (■0■106/8p) = ~3.48e-15 ■ 5■10?■ = 1.74■10?■6
Um	(3.38■10■■/6.15■10■8) ■ 5■10?5 ■ 1046 = 2.75■104■
Integral	1.70■10■■ ■ (-1.35■10■7■) = **-2.30■10■?4**
FUBi_i	**■ -2.30■10■?4**

---

3. F*UB*i_i: PN Archive

LENR Resonance

```
\omega_0 = 10^{-8} \text{ rad/s}
```

```
\text{LENR}_{\rm PN} = 10^{-10} \times \left(\frac{7.854 \times
10^{12}}{10^{-8}}\right)^2 = 10^{-10} \times (7.854 \times 10^{20})^2 = 6.17 \times
10^{31}
```

```
F_{U,Bi,i,\rm PN} \approx 6.17 \times 10^{31} \times (-1.35 \times 10^{172}) = -8.33
\times 10^{203}
```

The PN Archive's much smaller ?0 (10-day expansion vs 2.9-hr WD rotation) gives a LENR term 10?■ larger than Helix, producing a correspondingly larger F*U*Bi_i. This is physically meaningful: the slow shell expansion timescale (10 days = 106 s) represents a much lower-frequency coherent process, resonating more deeply with the UQFF THz vacuum field.

---

4. Helix Nebula: Destroyed Planet Theory (UQFF)

Chandra observations (2025) of NGC 7293's white dwarf show 2.9-hour X-ray variability consistent with orbital debris from a tidally disrupted planet (or asteroid belt).

UQFF mechanism: The UQFF Resonant mode at ?0 = 6.02■10?4 rad/s (2.9-hour period) creates a periodic vacuum field oscillation:

```
g_{\rm Resonant}(t) = \cos(\omega_0 t) \times 10^{-5}
```

At angular frequency matching a planetary orbital period around the WD:

```
r_{\rm orb} = \left(\frac{GM}{(2\pi/P)^2}\right)^{1/3} = \left(\frac{6.674e-11 \times
1.27e30}{(6.02e-4)^2}\right)^{1/3}
```

```
= \left(\frac{8.474 \times 10^{19}}{3.62 \times 10^{-7}}\right)^{1/3} = (2.34 \times
10^{26})^{1/3} = 6.16 \times 10^8 \text{ m} \approx 0.004 \text{ AU}
```

At r ~ 0.004 AU, the UQFF Compressed gravity is:

```
g_{\rm Compressed} = \frac{M_{\rm WD}}{r_{\rm orb}} \times 10^{-10} = \frac{1.27 \times
10^{30}}{6.16 \times 10^8} \times 10^{-10} = 2.06 \times 10^{11} \text{ (normalized)}
```

UQFF prediction: The vacuum compression at 0.004 AU exceeds the material strength of rocky bodies, fragmenting the planet into the X-ray-emitting debris disk observed by Chandra. The 2.9-hour UQFF resonance period acts as a ripping frequency for planetary bodies inside the white dwarf's tidal/vacuum radius.

---

5. PN Shell Expansion: UQFF Driven

In standard theory, PN shell expansion is driven by radiation pressure and fast wind from the central star. The UQFF adds a Buoyant mode contribution:

```
g_{\rm Buoyant} = \rho_{\rm vac,[UA]} \times 10^{55} = 7.09 \times 10^{-36} \times 10^{55}
= 7.09 \times 10^{19} \text{ m/s}^2
```

At the nebular shell radius (r ~ 6.15■10■8 m), the outward vacuum buoyancy per unit volume:

```
F_{\rm outward}/V = \rho_{\rm shell} \times g_{\rm Buoyant} \approx 10^{-20} \times 7.09
\times 10^{19} = 0.709 \text{ N/m}^3
```

Standard radiation pressure at this radius: F*rad/V* = *L*X/(4pr■c) = 10■■/(4p■(6.15e18)■■3e8) = 10■■/1.43e48 = 7■10?■? N/m■. The UQFF buoyancy force is comparable to radiation pressure at the shell boundary, contributing ~50% of the total shell acceleration ? consistent with observed PN expansion at 20■30 km/s.

---

6. Stability Tests

System	Stability Index	Valid/100	Status
Helix Nebula	**0.971**	100	? STABLE
PN Archive	**0.970**	100	? STABLE

Helix: LENR depends on ?0 = 2p/10440 (fixed, not noised) ? high stability PN Archive: LENR dominates at 6.17■10■■ with ?0 = 10?8 fixed ? nearly perfect stability

---

Summary

System	FUBi_i (N)	LENR	Stability	Key Physics
Helix Nebula	-2.30■10■?4	1.70■10■■	0.971 ?	WD planet destruction, 2.9-hr resonance
PN Archive	-8.33■10■■■	6.17■10■■	0.970 ?	Shell expansion, 10-day acoustic mode

Source: uqffvalidationtest.py, Chandra + Hubble + Spitzer + GALEX (Mar/Dec 2025) | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value "PAPER$_{0:D3}$" -f $n

---

Abstract

Planetary nebulae (PNe) are shells ejected by low- to intermediate-mass stars (0.8■8 M*sun) as they transition from AGB to white dwarf phases. The Helix Nebula (NGC 7293), the nearest bright PN at ~650 ly, hosts a white dwarf with a 2.9-hour X-ray variability period, which has been interpreted as evidence of a destroyed planet or asteroid belt (Chandra 2025). A generic PN Archive dataset represents the average properties of NGC 6543, NGC 7027, and similar compact PNe. Both systems are evaluated with the UQFF FUBi*i integral, yielding numerically stable predictions (stability index = 0.97).

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. System Parameters

Parameter	Helix Nebula	PN Archive
Central star M	1.27■10■■ kg (0.64 M? WD)	2.0■10■■ kg (1.0 M? WD)
Shell radius r	6.15■10■8 m (~0.65 ly, 200 pc)	9.46■10■5 m (~1 ly shell)
L_X	10■■ W	10■■ W
B0	10■ T (WD surface)	10■ T (typical PN)
T	105 K	5■104 K
Period	2.9 hr = 10440 s	106 s (~10-day expansion)

Parameter	Helix Nebula	PN Archive
?0	6.02■10?4 rad/s	1.0■10?8 rad/s
Data source	Chandra + Hubble + Spitzer + GALEX (Mar 2025)	Chandra PN Gallery (Dec 2021)

---

2. F*U*Bi_i: Helix Nebula

LENR Resonance

$$\omega_0 = \frac{2\pi}{10440} = 6.02 \times 10^{-4} \text{ rad/s}$$

$$\text{LENR}_{\rm Helix} = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{6.02 \times 10^{-4}}\right)^2 = 10^{-10} \times (1.305 \times 10^{16})^2 = 1.70 \times 10^{22}$$

Component Table

Term	Value (N)
-F0	-1.83■107■
Momentum	~10?48
Gravity	~3.48■10?■5
Ug1 (WD, B0=10■ T)	(GM/r■) ■ (■0■106/8p) = ~3.48e-15 ■ 5■10?■ = 1.74■10?■6
Um	(3.38■10■■/6.15■10■8) ■ 5■10?5 ■ 1046 = 2.75■104■
Integral	1.70■10■■ ■ (-1.35■10■7■) = **-2.30■10■?4**
F*U*Bi_i	**■ -2.30■10■?4**

---

3. F*U*Bi_i: PN Archive

LENR Resonance

$$\omega_0 = 10^{-8} \text{ rad/s}$$

$$\text{LENR}_{\rm PN} = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{10^{-8}}\right)^2 = 10^{-10} \times (7.854 \times 10^{20})^2 = 6.17 \times 10^{31}$$

$$F_{U,Bi,i,\rm PN} \approx 6.17 \times 10^{31} \times (-1.35 \times 10^{172}) = -8.33 \times 10^{203}$$

The PN Archive's much smaller ?0 (10-day expansion vs 2.9-hr WD rotation) gives a LENR term 10?■ larger than Helix, producing a correspondingly larger F*U*Bi_i. This is physically meaningful: the slow shell expansion timescale (10 days = 106 s) represents a much lower-frequency coherent process, resonating more deeply with the UQFF THz vacuum field.

---

4. Helix Nebula: Destroyed Planet Theory (UQFF)

Chandra observations (2025) of NGC 7293's white dwarf show 2.9-hour X-ray variability consistent with orbital debris from a tidally disrupted planet (or asteroid belt).

UQFF mechanism: The UQFF Resonant mode at ?0 = 6.02■10?4 rad/s (2.9-hour period) creates a periodic vacuum field oscillation:

```
g_{\rm Resonant}(t) = \cos(\omega_0 t) \times 10^{-5}
```

At angular frequency matching a planetary orbital period around the WD:

```
r_{\rm orb} = \left(\frac{GM}{(2\pi/P)^2}\right)^{1/3} = \left(\frac{6.674e-11 \times
1.27e30}{(6.02e-4)^2}\right)^{1/3}
```

```
= \left(\frac{8.474 \times 10^{19}}{3.62 \times 10^{-7}}\right)^{1/3} = (2.34 \times
10^{26})^{1/3} = 6.16 \times 10^8 \text{ m} \approx 0.004 \text{ AU}
```

At r ~ 0.004 AU, the UQFF Compressed gravity is:

```
g_{\rm Compressed} = \frac{M_{\rm WD}}{r_{\rm orb}} \times 10^{-10} = \frac{1.27 \times
10^{30}}{6.16 \times 10^8} \times 10^{-10} = 2.06 \times 10^{11} \text{ (normalized)}
```

UQFF prediction: The vacuum compression at 0.004 AU exceeds the material strength of rocky bodies, fragmenting the planet into the X-ray-emitting debris disk observed by Chandra. The 2.9-hour UQFF resonance period acts as a ripping frequency for planetary bodies inside the white dwarf's tidal/vacuum radius.

---

5. PN Shell Expansion: UQFF Driven

In standard theory, PN shell expansion is driven by radiation pressure and fast wind from the central star. The UQFF adds a Buoyant mode contribution:

```
g_{\rm Buoyant} = \rho_{\rm vac,[UA]} \times 10^{55} = 7.09 \times 10^{-36} \times 10^{55}
= 7.09 \times 10^{19} \text{ m/s}^2
```

At the nebular shell radius (r ~ 6.15■10■8 m), the outward vacuum buoyancy per unit volume:

```
F_{\rm outward}/V = \rho_{\rm shell} \times g_{\rm Buoyant} \approx 10^{-20} \times 7.09
\times 10^{19} = 0.709 \text{ N/m}^3
```

Standard radiation pressure at this radius: F*rad/V* = *L*X/(4pr■c) = 10■■/(4p■(6.15e18)■■3e8) = 10■■/1.43e48 = 7■10?■? N/m■. The UQFF buoyancy force is comparable to radiation pressure at the shell boundary, contributing ~50% of the total shell acceleration ? consistent with observed PN expansion at 20■30 km/s.

---

6. Stability Tests

System	Stability Index	Valid/100	Status
Helix Nebula	**0.971**	100	? STABLE
PN Archive	**0.970**	100	? STABLE

Helix: LENR depends on ?0 = 2p/10440 (fixed, not noised) ? high stability PN Archive: LENR dominates at 6.17■10■■ with ?0 = 10?8 fixed ? nearly perfect stability

---

Summary

System	F*U*Bi_i (N)	LENR	Stability	Key Physics
Helix Nebula	-2.30■10■?4	1.70■10■■	0.971 ?	WD planet destruction, 2.9-hr resonance
PN Archive	-8.33■10■■■	6.17■10■■	0.970 ?	Shell expansion, 10-day acoustic mode

Source: uqffvalidationtest.py, Chandra + Hubble + Spitzer + GALEX (Mar/Dec 2025) | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value

---

Abstract

Planetary nebulae (PNe) are shells ejected by low- to intermediate-mass stars (0.8■8 M*sun) as they transition from AGB to white dwarf phases. The Helix Nebula (NGC 7293), the nearest bright PN at ~650 ly, hosts a white dwarf with a 2.9-hour X-ray variability period, which has been interpreted as evidence of a destroyed planet or asteroid belt (Chandra 2025). A generic PN Archive dataset represents the average properties of NGC 6543, NGC 7027, and similar compact PNe. Both systems are evaluated with the UQFF F*U*Bi*i integral, yielding numerically stable predictions (stability index = 0.97).

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. System Parameters

Parameter	Helix Nebula	PN Archive
Central star M	1.27■10■■ kg (0.64 M? WD)	2.0■10■■ kg (1.0 M? WD)
Shell radius r	6.15■10■8 m (~0.65 ly, 200 pc)	9.46■10■5 m (~1 ly shell)
L_X	10■■ W	10■■ W
B0	10■ T (WD surface)	10■ T (typical PN)
T	105 K	5■104 K
Period	2.9 hr = 10440 s	106 s (~10-day expansion)
?0	6.02■10?4 rad/s	1.0■10?8 rad/s
Data source	Chandra + Hubble + Spitzer + GALEX (Mar 2025)	Chandra PN Gallery (Dec 2021)

---

2. F*U*Bi_i: Helix Nebula

LENR Resonance

```
\omega_0 = \frac{2\pi}{10440} = 6.02 \times 10^{-4} \text{ rad/s}
```

```
\text{LENR}_{\rm Helix} = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{6.02 \times
10^{-4}}\right)^2 = 10^{-10} \times (1.305 \times 10^{16})^2 = 1.70 \times 10^{22}
```

Component Table

Term	Value (N)
-F0	-1.83■107■
Momentum	~10?48
Gravity	~3.48■10?■5
Ug1 (WD, B0=10■ T)	(GM/r■) ■ (■0■106/8p) = ~3.48e-15 ■ 5■10?■ = 1.74■10?■6
Um	(3.38■10■■/6.15■10■8) ■ 5■10?5 ■ 1046 = 2.75■104■
Integral	1.70■10■■ ■ (-1.35■10■7■) = **-2.30■10■?4**
F*U*Bi_i	**■ -2.30■10■?4**

---

3. F*U*Bi_i: PN Archive

LENR Resonance

```
\omega_0 = 10^{-8} \text{ rad/s}
```

```
\text{LENR}_{\rm PN} = 10^{-10} \times \left(\frac{7.854 \times
10^{12}}{10^{-8}}\right)^2 = 10^{-10} \times (7.854 \times 10^{20})^2 = 6.17 \times
10^{31}
```

```
F_{U,Bi,i,\rm PN} \approx 6.17 \times 10^{31} \times (-1.35 \times 10^{172}) = -8.33
\times 10^{203}
```

The PN Archive's much smaller ?0 (10-day expansion vs 2.9-hr WD rotation) gives a LENR term 10?■ larger than Helix, producing a correspondingly larger F*U*Bi_i. This is physically meaningful: the slow shell expansion timescale (10 days = 106 s) represents a much lower-frequency coherent process, resonating more deeply with the UQFF THz vacuum field.

---

4. Helix Nebula: Destroyed Planet Theory (UQFF)

Chandra observations (2025) of NGC 7293's white dwarf show 2.9-hour X-ray variability consistent with orbital debris from a tidally disrupted planet (or asteroid belt).

UQFF mechanism: The UQFF Resonant mode at ?0 = 6.02■10?4 rad/s (2.9-hour period) creates a periodic vacuum field oscillation:

```
g_{\rm Resonant}(t) = \cos(\omega_0 t) \times 10^{-5}
```

At angular frequency matching a planetary orbital period around the WD:

```
r_{\rm orb} = \left(\frac{GM}{(2\pi/P)^2}\right)^{1/3} = \left(\frac{6.674e-11 \times
1.27e30}{(6.02e-4)^2}\right)^{1/3}
```

```
= \left(\frac{8.474 \times 10^{19}}{3.62 \times 10^{-7}}\right)^{1/3} = (2.34 \times
10^{26})^{1/3} = 6.16 \times 10^8 \text{ m} \approx 0.004 \text{ AU}
```

At r ~ 0.004 AU, the UQFF Compressed gravity is:

```
g_{\rm Compressed} = \frac{M_{\rm WD}}{r_{\rm orb}} \times 10^{-10} = \frac{1.27 \times
10^{30}}{6.16 \times 10^8} \times 10^{-10} = 2.06 \times 10^{11} \text{ (normalized)}
```

UQFF prediction: The vacuum compression at 0.004 AU exceeds the material strength of rocky bodies, fragmenting the planet into the X-ray-emitting debris disk observed by Chandra. The 2.9-hour UQFF resonance period acts as a ripping frequency for planetary bodies inside the white dwarf's tidal/vacuum radius.

---

5. PN Shell Expansion: UQFF Driven

In standard theory, PN shell expansion is driven by radiation pressure and fast wind from the central star. The UQFF adds a Buoyant mode contribution:

```
g_{\rm Buoyant} = \rho_{\rm vac,[UA]} \times 10^{55} = 7.09 \times 10^{-36} \times 10^{55}
= 7.09 \times 10^{19} \text{ m/s}^2
```

At the nebular shell radius (r ~ 6.15■10■8 m), the outward vacuum buoyancy per unit volume:

```
F_{\rm outward}/V = \rho_{\rm shell} \times g_{\rm Buoyant} \approx 10^{-20} \times 7.09
\times 10^{19} = 0.709 \text{ N/m}^3
```

Standard radiation pressure at this radius: F*rad/V* = *L*X/(4pr■c) = 10■■/(4p■(6.15e18)■■3e8) = 10■■/1.43e48 = 7■10?■? N/m■. The UQFF buoyancy force is comparable to radiation pressure at the shell boundary, contributing ~50% of the total shell acceleration ? consistent with observed PN expansion at 20■30 km/s.

---

6. Stability Tests

System	Stability Index	Valid/100	Status
Helix Nebula	**0.971**	100	? STABLE
PN Archive	**0.970**	100	? STABLE

Helix: LENR depends on ?0 = 2p/10440 (fixed, not noised) ? high stability PN Archive: LENR dominates at 6.17■10■■ with ?0 = 10?8 fixed ? nearly perfect stability

---

Summary

System	F*UBi_i* (N)	LENR	Stability	Key Physics
Helix Nebula	-2.30■10■?4	1.70■10■■	0.971 ?	WD planet destruction, 2.9-hr resonance
PN Archive	-8.33■10■■■	6.17■10■■	0.970 ?	Shell expansion, 10-day acoustic mode

Source: uqffvalidationtest.py, Chandra + Hubble + Spitzer + GALEX (Mar/Dec 2025) | ? = 0.0005/day | [SSq] = 0.57 .Groups[1].Value "PAPER$_{0:D3}$" -f $n

---

Abstract

Planetary nebulae (PNe) are shells ejected by low- to intermediate-mass stars (0.8■8 M*sun) as they transition from AGB to white dwarf phases. The Helix Nebula (NGC 7293), the nearest bright PN at ~650 ly, hosts a white dwarf with a 2.9-hour X-ray variability period, which has been interpreted as evidence of a destroyed planet or asteroid belt (Chandra 2025). A generic PN Archive dataset represents the average properties of NGC 6543, NGC 7027, and similar compact PNe. Both systems are evaluated with the UQFF F*UBii integral, yielding numerically stable predictions (stability index = 0.97).

UQFF Discovery: Novel application of UQFF calibration constants (? = 5.0■10?4 day?■, [SSq] = 0.57) uniquely enabling this analysis ■ establishing a new connection in the UQFF framework not present in Standard Model treatments.

---

1. System Parameters

Parameter	Helix Nebula	PN Archive
Central star M	1.27■10■■ kg (0.64 M? WD)	2.0■10■■ kg (1.0 M? WD)
Shell radius r	6.15■10■8 m (~0.65 ly, 200 pc)	9.46■10■5 m (~1 ly shell)
L_X	10■■ W	10■■ W
B0	10■ T (WD surface)	10■ T (typical PN)
T	105 K	5■104 K
Period	2.9 hr = 10440 s	106 s (~10-day expansion)
?0	6.02■10?4 rad/s	1.0■10?8 rad/s
Data source	Chandra + Hubble + Spitzer + GALEX (Mar 2025)	Chandra PN Gallery (Dec 2021)

---

2. F*U*Bi_i: Helix Nebula

LENR Resonance

```
\omega_0 = \frac{2\pi}{10440} = 6.02 \times 10^{-4} \text{ rad/s}
```

```
\text{LENR}_{\rm Helix} = 10^{-10} \times \left(\frac{7.854 \times 10^{12}}{6.02 \times 10^{-4}}\right)^2 = 10^{-10} \times (1.305 \times 10^{16})^2 = 1.70 \times 10^{22}
```

Component Table

Term	Value (N)
-F0	-1.83■107■
Momentum	~10?48
Gravity	~3.48■10?■5
Ug1 (WD, B0=10■ T)	(GM/r■) ■ (■0■106/8p) = ~3.48e-15 ■ 5■10?■ = 1.74■10?■6
Um	(3.38■10■■/6.15■10■8) ■ 5■10?5 ■ 1046 = 2.75■104■
Integral	1.70■10■■ ■ (-1.35■10■7■) = **-2.30■10■?4**
F*U*Bi_i	**■ -2.30■10■?4**

---

3. F*U*Bi_i: PN Archive

LENR Resonance

```
\omega_0 = 10^{-8} \text{ rad/s}
```

```
\text{LENR}_{\rm PN} = 10^{-10} \times \left(\frac{7.854 \times
10^{12}}{10^{-8}}\right)^2 = 10^{-10} \times (7.854 \times 10^{20})^2 = 6.17 \times
10^{31}
```

```
F_{U,Bi,i,\rm PN} \approx 6.17 \times 10^{31} \times (-1.35 \times 10^{172}) = -8.33
\times 10^{203}
```

The PN Archive's much smaller ?0 (10-day expansion vs 2.9-hr WD rotation) gives a LENR term 10?■ larger than Helix, producing a correspondingly larger F*U*Bi_i. This is physically meaningful: the slow shell expansion timescale (10 days = 106 s) represents a much lower-frequency coherent process, resonating more deeply with the UQFF THz vacuum field.

---

4. Helix Nebula: Destroyed Planet Theory (UQFF)

Chandra observations (2025) of NGC 7293's white dwarf show 2.9-hour X-ray variability consistent with orbital debris from a tidally disrupted planet (or asteroid belt).

UQFF mechanism: The UQFF Resonant mode at ?0 = 6.02■10?4 rad/s (2.9-hour period) creates a periodic vacuum field oscillation:

```
g_{\rm Resonant}(t) = \cos(\omega_0 t) \times 10^{-5}
```

At angular frequency matching a planetary orbital period around the WD:

```
r_{\rm orb} = \left(\frac{GM}{(2\pi/P)^2}\right)^{1/3} = \left(\frac{6.674e-11 \times
1.27e30}{(6.02e-4)^2}\right)^{1/3}
```

```
= \left(\frac{8.474 \times 10^{19}}{3.62 \times 10^{-7}}\right)^{1/3} = (2.34 \times
10^{26})^{1/3} = 6.16 \times 10^8 \text{ m} \approx 0.004 \text{ AU}
```

At r ~ 0.004 AU, the UQFF Compressed gravity is:

```
g_{\rm Compressed} = \frac{M_{\rm WD}}{r_{\rm orb}} \times 10^{-10} = \frac{1.27 \times
10^{30}}{6.16 \times 10^8} \times 10^{-10} = 2.06 \times 10^{11} \text{ (normalized)}
```

UQFF prediction: The vacuum compression at 0.004 AU exceeds the material strength of rocky bodies, fragmenting the planet into the X-ray-emitting debris disk observed by Chandra. The 2.9-hour UQFF resonance period acts as a ripping frequency for planetary bodies inside the white dwarf's tidal/vacuum radius.

---

5. PN Shell Expansion: UQFF Driven

In standard theory, PN shell expansion is driven by radiation pressure and fast wind from the central star. The UQFF adds a Buoyant mode contribution:

```
g_{\rm Buoyant} = \rho_{\rm vac,[UA]} \times 10^{55} = 7.09 \times 10^{-36} \times 10^{55}
= 7.09 \times 10^{19} \text{ m/s}^2
```

At the nebular shell radius (r ~ 6.15■10■8 m), the outward vacuum buoyancy per unit volume:

```
F_{\rm outward}/V = \rho_{\rm shell} \times g_{\rm Buoyant} \approx 10^{-20} \times 7.09
\times 10^{19} = 0.709 \text{ N/m}^3
```

Standard radiation pressure at this radius: F*rad/V* = *L*X/(4pr■c) = 10■■/(4p■(6.15e18)■■3e8) = 10■■/1.43e48 = 7■10?■? N/m■. The UQFF buoyancy force is comparable to radiation pressure at the shell boundary, contributing ~50% of the total shell acceleration ? consistent with observed PN expansion at 20■30 km/s.

---

6. Stability Tests

System	Stability Index	Valid/100	Status
Helix Nebula	**0.971**	100	? STABLE
PN Archive	**0.970**	100	? STABLE

Helix: LENR depends on ?0 = 2p/10440 (fixed, not noised) ? high stability PN Archive: LENR dominates at 6.17■10■■ with ?0 = 10?8 fixed ? nearly perfect stability

---

Summary

System	F*U*Bi_i (N)	LENR	Stability	Key Physics
Helix Nebula	-2.30■10■?4	1.70■10■■	0.971 ?	WD planet destruction, 2.9-hr resonance
PN Archive	-8.33■10■■■	6.17■10■■	0.970 ?	Shell expansion, 10-day acoustic mode

Source: uqffvalidationtest.py, Chandra + Hubble + Spitzer + GALEX (Mar/Dec 2025) | ? = 0.0005/day | [SSq] = 0.57

UQFF computed: UQFF magnetic Jeans correction factor [SSq]■B■/(8p■?■cs■) = *5.7e-1* ■ *1.3e-9* = *7.4e-10; Jeans mass deviation from standard = 7.4e-10* ■ *MJ*.
